Sutanu Bhattacharya, Ph.D.
Assistant Professor
Department of Computer Science
Auburn University at Montgomery
USA

URL: http://sutanubh1.github.io/
Mobile: +1 334 444 5256
E-Mail: sbhatta4@aum.edu

Education

Ph.D. in Computer Science and Software Engineering — Summer 2021
Auburn University, USA
Advisor: Dr. Debswapna Bhattacharya (Now at Virginia Tech)
Dissertation Title: New computational and data-driven methods for protein homology modeling

M.Tech in Distributed and Mobile Computing — June 2014
Jadavpur University, India

B.Tech in Information Technology — June 2011
Bengal Institute of Technology, India

Employment

Assistant Professor, Department of Computer Science, Auburn University at Montgomery, USA, 16 August 2022 - ongoing.

Assistant Professor, Department of Computer Science, Florida Polytechnic University, USA, September 2021 – 15 August 2022.

Assistant Professor, Department of Computer Science and Engineering, Chaibasa Engineering College, India, July 2014 – Aug 2017.

Research Interest

My research lies at the intersection of **computational biology and artificial intelligence (AI)**, where I develop AI-driven methods and computational algorithms to advance structural bioinformatics. A central goal of my work is to develop new approaches for protein sequence alignment, remote homology detection, and functional inference, areas that remain critical despite recent breakthroughs like AlphaFold2. While these advances have transformed the field, important challenges remain, particularly in detecting similarities among distant-homology proteins, analyzing large-scale metagenomic datasets, and integrating sequence information with functional and structural insights. My research addresses these gaps by combining embedding-based representations with scalable algorithms and deep learning models, creating solutions that are both scientifically rigorous and practical for large-scale applications. In the long term, my goal is to develop AI-driven tools that integrate sequence, structure, and function analysis, contributing to advances in structural biology, drug discovery, and genomics.

In addition, I also contribute to collaborative projects that extend my AI expertise into other domains, including multilingual sentiment analysis and the NSF ExpandAI project, where I serve as Co-PI and lead the machine learning (ML)-Bot initiative to enhance student research engagement.

Peer-reviewed publications (+ undergraduate student, ++ graduate student, * as corresponding author; Selected journals include Bioinformatics (ranked #1 in Bioinformatics & Computational Biology), PLOS Computational Biology (ranked #3), Proteins: Structure, Function, and Bioinformatics (top 10 in Proteomics), and Nature Scientific Reports (top 10 in Life Sciences & Earth Sciences), based on Google Scholar rankings.)

In-preparation

(27) S. Danwada++, P. Udomprasert+, **S. Bhattacharya***, "Unsupervised Clustering of Protein Language Model Embeddings for Homology Detection", 2025. (*Selected for Lightning Talk and Poster at 21st MCBIOS 2025 Conference, Utah*)

(26) S. P. Pallati++, S. Danwada++, N. Raychawdhary, **S. Bhattacharya***, "Deep Learning-Powered Protein Similarity Search: A Scalable Sequence-Based Approach", 2025. (*Selected for Poster at 21st MCBIOS 2025 Conference, Utah*)

Under review

(25) R. Spicer++, N. Raychawdhary, C. Seals, **S. Bhattacharya***, "Embedding-Based Sequence Alignment Using Clustering and Double Dynamic Programming", Nature Scientific Reports, 2025. (R. Spicer is secured First Position in Young Research Excellence Award (graduate student category) at the 21st MCBIOS 2025 Conference; College Award at Auburn University Research Symposium 2025). (*Impact Factor: 3.9*)

(24) S. Mallela, A. Gates, **S. Bhattacharya**, B. Okeke, O. Kursun, "AI-Based Detection of Coliform Colonies Using CNN Transfer Learning for Application to Cultured Plate Analysis in Water Quality Research", Computational and Structural Biotechnology Journal (CSBJ), 2025. (*Impact Factor: 4.1*)

Book Chapter

(23) **S. Bhattacharya,** R. Roche, M. H. Shuvo, B. Moussad, D. Bhattacharya, "Contact-assisted threading in low-homology protein modeling", Methods in Molecular Biology, by **Springer Nature**, 2023. *(Impact Factor: 10.71)*

Journals

(22) N. Raychawdhary, **S. Bhattacharya**, G. Dozier, C. Seals, "Empowering Sentiment Analysis in African Low-Resource Languages through Transformer Models and Strategic Language Selection", **IEEE ACCESS**, 2025. *(Impact Factor: 3.6)*

(21) N. Raychawdhary, **S. Bhattacharya**, G. Dozier, C. Seals, "Teaching NLP and Machine Learning Through Case Studies Using Interactive Environments", **Journal of the Consortium for Computing Sciences in Colleges**.

(20) K. Baker++, N. Hughes+, **S. Bhattacharya***, "An interactive visualization tool for educational outreach in protein contact map overlap analysis", **Frontiers in Bioinformatics**, 4, 2024. (*Selected for an oral presentation and Best Poster Award (3rd) at the 19th Annual MCBIOS Conference, Dallas, USA, March 15-17, 2023*). *(Impact Factor: 3.9)*

(19) R. Roche, **S. Bhattacharya**, M. H. Shuvo, D. Bhattacharya, "rrQNet: protein contact map quality estimation by deep evolutionary reconciliation", **Proteins: Structure, Function, and**

Bioinformatics, https://doi.org/10.1002/prot.26394, (2022). *(Impact Factor: 2.8)*

(18) **S. Bhattacharya**, R. Roche, B. Moussad, D. Bhattacharya, "DisCovER: distance- and orientation-based covariational threading for weakly homologous proteins", **Proteins: Structure, Function, and Bioinformatics**, (2021). (**First method to utilize inter-residue orientations to boost protein threading performance**, Fellowship Award for 29th ISMB/ECCB). *(Impact Factor: 2.8)*

(17) Kryshtafovych et al. (2021). Modeling SARS-CoV2 proteins in the CASP-commons experiment. ***Proteins: Structure, Function, and Bioinformatics***, (2021). *(Impact Factor: 2.8)*

(16) **S. Bhattacharya,** R. Roche, M. H. Shuvo, D. Bhattacharya, "Recent advances in protein homology detection propelled by inter-residue interaction map threading", **Frontiers in Molecular Biosciences**, 8, 377 (2021). *(Impact Factor: 4.0)*

(15) R. Roche, **S. Bhattacharya**, D. Bhattacharya, "Hybridized distance- and contact-based hierarchical structure modeling for folding soluble and membrane proteins", **PLOS Computational Biology, 17(2): e1008753**, (2021). (Selected as Highlight Talk at ACM-BCB). *(Impact Factor: 3.6)*

(14) A. McGehee, **S. Bhattacharya**, R. Roche, D. Bhattacharya, "PolyFold: An interactive visual simulator for distance-based protein folding", **PLOS ONE**, 15(12): e0243331 (2020). (Best Poster Award at ACM-BCB). *(Impact Factor: 2.6)*

(13) M. H. Shuvo, **S. Bhattacharya**, D. Bhattacharya, "QDeep: distance-based protein model quality estimation by residue-level ensemble error classifications using stacked deep residual neural networks", **ISMB Proceedings**, **Bioinformatics**, 36(S1): i285-i291 (2020). *(Impact Factor: 5.4)*

(12) **S. Bhattacharya**, D. Bhattacharya, "Evaluating the significance of contact maps in low-homology protein modeling using contact-assisted threading", **Nature Scientific Reports**, 10(1), 1-13 (2020). *(Impact Factor: 3.9)*

(11) **S. Bhattacharya**, D. Bhattacharya, "Does inclusion of residue-residue contact information boost protein threading?", **Proteins: Structure, Function, and Bioinformatics**, 87(7): 596-606 (2019). (Selected as Front Cover Article, Highlight Talk at ACM-BCB, Top Downloaded Paper of 2018-2019 by WILEY). *(Impact Factor: 2.8)*

Conferences

(10) N. Raychawdhary, **S. Bhattacharya**, G. Dozier, C. Seals, "Empowering Undergraduates with NLP: Integrative Methods for Deepening Understanding through Visualization and Case Studies", **ASEE Southeast Conference, USA,** 2025.

(9) N. Raychawdhary, A. Das, **S. Bhattacharya**, G. Dozier, C. Seals, "Optimizing Multilingual Sentiment Analysis in Low-Resource Languages with Adaptive Pretraining and Strategic Language Selection", **IEEE ICMI**, USA, 2024.

(8) N. Raychawdhary, A. Das, **S. Bhattacharya**, G. Dozier, C. Seals, "Enhancing Monolingual Sentiment Classification: Pioneering Strategies in Tailored Language Training and Analytical Techniques", **IEEE ICMI**, USA, 2024.

(7) N. Raychawdhary, A. Das, **S. Bhattacharya**, G. Dozier, C. Seals, "Enhancing Sentiment Analysis in Amharic: Leveraging Transformer-Based Language Model for Low-Resource African Languages", **IEEE SoutheastCon**, USA, 2024.

(6) N. Raychawdhary, N. Hughes+, **S. Bhattacharya**, G. Dozier, C. Seals, "A Transformer-Based Language Model for Sentiment Classification and Cross-Linguistic Generalization: Empowering Low-Resource African Languages", **IEEE AIBThings**, USA, 2023.

Workshop

(5) N. Hughes+, K. Baker++, Aditya Singh+, Aryavardhan Singh+, T. Dauda+, **S. Bhattacharya***, "Bhattacharya_Lab at SemEval-2023 task 12: A Transformer-based language model for sentiment classification for low-resource African languages: Nigerian Pidgin and Yoruba", In Proceedings of the 17th International Workshop on Semantic Evaluation (**SemEval-2023**), pages 1502-1507, Toronto, Canada, July 2023. Association for Computational Linguistics. (Ranked 1 out of 33 groups for Nigerian Pidgin and Ranked 5 out of 33 groups for Yoruba).

Abstracts

(4) R. Roche, **S. Bhattacharya**, D. Bhattacharya, "Hybridized distance- and contact-based hierarchical structure modeling for folding soluble and membrane proteins", **BCB '21**: Proceeding of the 12th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics, USA, August 2021, Pages 1 (2021).

(3) **S. Bhattacharya**, D. Bhattacharya, "How Effective is Contact-assisted protein threading?", **BCB'19**: Proceedings of the 10th **ACM** International Conference on Bioinformatics, Computational Biology and Health Informatics, September 2019, Pages 553 (2019).

(2) **S. Bhattacharya**, D. Bhattacharya, "Contact-assisted protein threading: an evolving new direction", **BCB'19**: Proceedings of the 10th **ACM** International Conference on Bioinformatics, Computational Biology and Health Informatics, September 2019, Pages 536 (2019).

Journal Cover Image

(1) **S. Bhattacharya**, D. Bhattacharya, Cover Image, Proteins: Structure, Function, and Bioinformatics, 87(7): C1-C1 (2019).

Conference Talks / Invited Talks

(7) "Unsupervised Clustering of Protein Language Model Embeddings for Homology Detection", 21st **MCBIOS** Conference, Salt Lake City, Utah, USA, 2025.

(6) "A real-time protein folding program with enhanced alignment capability", at California State University, San Marcos, 2023.

(5) "GoFold: a real-time protein folding program", 19th **MCBIOS** conference, Dallas, USA, 2023.

(4) "A distance- and orientation-based covariational protein threading", at Florida Polytechnic University, USA, 2022.

(3) "DisCovER: distance- and orientation-based covariational threading for weakly homologous proteins", 29th **ISMB/ECCB** conference, Virtual Event, July 25-30, 2021. (Fellowship Award)

(2) "How Effective is Contact-assisted Protein Threading?", 10th **ACM-BCB** Conference, Niagara Falls, NY, Sept 7-10, 2019. (Highlight Talk)

(1) "Does inclusion of residue-residue contact information boost protein threading?", 16th Annual **MCBIOS** Conference, Birmingham, USA, March 28- 30, 2019 (Won 2nd place in student oral presentation).

Posters

As Adviser (+ undergraduate student, ++ graduate student)

(12) P. Udomprasert+, **S. Bhattacharya**, "Unsupervised Clustering of Protein Language Model Embeddings for Homology Detection", 21st **MCBIOS** Conference, Salt Lake City, Utah, USA, 2025.

(11) S. P. Pallati++, R. Spicer++, **S. Bhattacharya**, "Deep Learning-Powered Protein Similarity Search: A Scalable Sequence-Based Approach", 21st **MCBIOS** Conference, Salt Lake City, Utah, USA, 2025.

(10) R. Spicer+, N. Hughes+, **S. Bhattacharya,** "Leveraging Transformer Models for Sentiment Analysis in African Languages: A Cross-Linguistic Approach", **ACM CMD-IT TAPIA** Conference, San Diego, USA, 2024.

(9) R. Spicer+, N. Hughes+, **S. Bhattacharya,** "Transformer-Based Approaches to Sentiment Analysis in African Languages", **Auburn University Research Symposium**, USA, 2024. (College Award for AUM)

(8) N. Hughes+, **S. Bhattacharya,** "AfriSenti-SemEval: Exploring the viability of pre-trained multilingual language models for monolingual sentiment classification for low-resources African languages", **ACM CMD-IT TAPIA** Conference, Dallas, USA, 2023. (3rd prize in Best Poster Award category)

(7) K. Baker++, N. Hughes+, **S. Bhattacharya**, "GoFold: a real-time protein folding program", 19th **MCBIOS** conference, Dallas, USA, 2023. (3rd prize in Best Poster Award category)

(6) N. Hughes+, K. Baker++, A. Singh+, A. Singh+, T. Dauda+, **S. Bhattacharya,** **"**Bhattacharya_Lab at SemEval-2023 task 12: A Transformer-based language model for sentiment classification for low-resource African languages: Nigerian Pidgin and Yoruba**",** 3rd **Annual Celebration of Undergraduate Research and Creative Activity,** Auburn University at Montgomery, 2023.

As Presenter

(5) **S. Bhattacharya**, R. Roche, D. Bhattacharya, "DisCovER: distance- and orientation-based covariational threading for weakly homologous proteins", 29th **ISMB/ECCB** conference, Virtual Event, July 25-30, 2021. (Fellowship Award)

(4) A. McGehee, **S. Bhattacharya**, R. Roche, D. Bhattacharya, "PolyFold: An interactive visual

simulator for distance-based protein folding", 11th **ACM-BCB** Conference, Virtual Event, Sept 21-24, 2020. (Best Poster Award)

(3) **S. Bhattacharya**, D. Bhattacharya, "Contact-assisted protein threading: an evolving new direction", 2019 Graduate Engineering Research Showcase, Auburn University, USA, November 7, 2019.

(2) **S. Bhattacharya**, D. Bhattacharya, "Contact-assisted protein threading: an evolving new direction", 10th **ACM-BCB** Conference, Niagara Falls, NY, Sept 7-10, 2019.

(1) **S. Bhattacharya**, D. Bhattacharya, "Does inclusion of residue-residue contact information boost protein threading?", 16th Annual **MCBIOS** Conference, Birmingham, USA, March 28-30, 2019.

Grants

Awarded Grants

(10) **NSF Expand AI**, "CAP: Developing a Machine Learning Lab for AI-Powered Training and Research", 2024-26, Amount: $400, 000 (role: Co-PI). Major award supporting development of GPU infrastructure and AI research capacity at AUM. Leading the ML-Bot initiative, a key deliverable designed to enhance AI-powered student research engagement.

(9) **Google Research Cloud Award**, 2025-26, Amount: $5000 of Cloud Resources (role: PI).

(8) **UGRCA Award**, Auburn University at Montgomery, 2025-26, Amount: $2057 (role: PI).

(7) **GSRAC Conference Travel Award**, Auburn University at Montgomery, 2025, Amount: $5984.46 (role: PI)

(6) **GIA Award**, Auburn University at Montgomery, 2022-24, Amount: $7500 (role: PI).

(5) Start-up Grant, Auburn University at Montgomery, 2022-25, Amount: $5000 (role: PI)

(4) Summer Research Grant, Florida Polytechnic University, 2022 (role: PI).

Proposals in Preparation

(12) **NSF**, "Zero-Shot Prediction of Protein–Protein Interaction Sites Using Language Model Embeddings", Amount: $415,000 (role: PI).

(11) **NSF**, "LLM-Driven Tutors for Teaching Bioinformatics Algorithms", Amount: $300,000 (role: PI).

Submitted Proposals (Not Funded)

(3) **NSF** (2025), "CISE MSI: RCBP: III: Building Research Capacity in AI-Driven Bioinformatics at Auburn University at Montgomery", Amount: $400,000 (role: PI).

(2) **Google Research Scholar Program** (2025), "AI-Powered Protein Sequence Alignment in the Twilight Zone", Amount: $50, 000 (role: PI).

(1) **NSF** (2023), "CAP: Cortex-Inspired Cross-Modal Deep Learning and Application to Bioimage Analysis", Amount: $399, 978 (role: Co-PI).

Honors and Awards

(12) **Departmental Outstanding Faculty Teaching Award 2025**, Auburn University at Montgomery, 2025.

(11) **GSRAC Conference travel award** from AUM, 21st Annual MCBIOS Conference, Utah, 2025.

(10) **Faculty Award**, ACM CMD/IT TAPIA Conference, San Diego, USA, 2024.

(9) **Ranked 1** (out of 33 teams)**,** AfriSenti-SemEval: Sentiment Analysis for African Language (Task 12), 17th International Workshop on Semantic Evaluation, Canada, 2023.

(8) **Highlight Talk,** 12th ACM-BCB conference (Virtual), Aug 1-4, 2021.

(7) **Fellowship award,** ISMB/ECCB 2021 conference (Virtual), July 25 – 30, 2021.

(6) **Best Poster** (1st place) award, 11th ACM-BCB conference (Virtual), Sept 21-24, 2020.

(5) **Highlight Talk,** 10th ACM-BCB conference, Niagara Falls, NY, Sept 7-10, 2019.

(4) **Front Cover Article,** July 2019 issue of Proteins: Structure, Function, and Bioinformatics journal.

(3) **Young Research Excellence Award** (2nd place), 16th Annual MCBIOS Conference, Birmingham, USA, 2019.

(2) **Travel Award**, 16th Annual MCBIOS Conference, Birmingham, USA, 2019.

(1) **National Scholarship**, Jadavpur University, India, 2012 – 2014.

Students' Honors and Awards (+ undergraduate student, ++ graduate student)

(16) Sai Prashanthi Pallati++, AUM, **Travel Award**, ACM CMD/IT TAPIA conference at Dallas (USA), 2025.

(15) William Rochelle+, AUM, STEM scholarship at AUM, 2025.

(14) Sai Prashanthi Pallati++, AUM, **Graduate Poster Presentation Award** (1st), COS Student Research Symposium at AUM, 2025.

(13) Robert Spicer++, AUM, **Graduate Oral Presentation Award** (1st) and **People's Choice Awards**, COS Student Research Symposium at AUM, 2025.

(12) Robert Spicer++, AUM, **Travel Award**, CRA-WP Grad Cohort for Inclusion, Diversity, Equity, Accessibility, and Leadership Skills (IDEALS) Workshop, Denver, Colorado, 2025.

(11) Robert Spicer++, AUM, **First Position**, **Young Research Excellence Award** (Graduate Student Category), 21st Annual MCBIOS Conference, Utah, 2025. Despite being a master's

student, Robert stood out among other finalists, who were PhD students from R1 research universities such as the University of Pittsburgh, University of Utah, and Ohio State University.

(10) Robert Spicer[++], AUM, **College Award** for Graduate Presentation for Auburn University at Montgomery, Auburn Research Symposium at Auburn University, 2025.

(9) Sai Prashanthi Pallati[++], AUM, **GSRAC Conference travel award** from AUM, 21st Annual MCBIOS Conference, Utah, 2025.

(8) Robert Spicer[++], AUM, **GSRAC Conference travel award** from AUM, 21st Annual MCBIOS Conference, Utah, 2025.

(7) Robert Spicer[+], AUM, **Travel Award**, ACM CMD/IT TAPIA conference at San Diego, USA, 2024.

(6) William Rochelle[+], AUM, **2024 GABBR LSAMP summer research internship** to conduct summer research under my direction, Summer 2024.

(5) Robert Spicer[+], AUM, **College Award** for Undergraduate Poster Presentation for Auburn University at Montgomery, Auburn Research Symposium at Auburn University, 2024.

(4) Nathaniel Hughes[+], AUM, **Outstanding Student Award for Computer Science Department,** 2024.

(3) Kevan Baker[++], Florida Polytechnic University, **MS thesis**, Spring 2023. Joined Auburn University as a PhD student.

(2) Nathaneil Hughes[+], AUM, **Best Poster Award** (3rd) and **Travel Award** at ACM CMD/IT TAPIA conference at Dallas, USA, 2023.

(1) Kevan Baker[++], Florida Polytechnic University, **Best Poster Award** (3rd place) and **Travel Award**, 19th Annual MCBIOS Conference, Dallas, USA, 2023.

Teaching, Mentoring, and Advising Activities

Teaching Experience

Auburn University at Montgomery (Fall 2022 – now)

Academic Term	**Course**	**Enrollment**
Fall 2025	CSCI 6000 Data Structure & Fund Algorithms Design	21
	BCMB 6030: Principles of Bioinformatics	5
	CSCI 6970 SpTp: Natural Language Processing	18
Summer 2025	CSCI 6000 Data Structure & Fund Algorithms Design	13
Spring 2025	CSCI 6000 Data Structure & Fund Algorithms Design	63
Fall 2024	CSCI 6000 Data Structure & Fund Algorithms Design	53
	CSCI 3400 Data Structures	18
Summer 2024	CSCI 6000 Data Structure & Fund Algorithms Design *(Online)*	55
Spring 2024	CSCI 6000 Data Structure & Fund	174

	Algorithms Design	
	CSCI 3400 Data Structures	20
	CSCI 3600 Fund Algorithm Design and Analysis	12
Fall 2023	CSCI 6000 Data Structure & Fund Algorithms Design	142
	CSCI 3400 Data Structures	11
	CSCI 3600 Fund Algorithm Design and Analysis	14
Summer 2023	CSCI 6170 Advanced Network Systems *(Online)*	74
Spring 2023	CSCI 6170 Advanced Network Systems	33
	CSCI 3400 Data Structures	13
	CSCI 3600 Fund Algorithm Design and Analysis	17
Fall 2022	CSCI 6170 Advanced Network Systems	100
	CSCI 3400 Data Structures	17
	CSCI 3600 Fund Algorithm Design and Analysis	18
	CSCI 2200 Discrete Structures	11
	CSCI 4350 Network Systems *(Online)*	8

Florida Polytechnic University (Fall 2021 – Spring 2022)

Academic Term	**Course**	**Enrollment**
Spring 2022	COP 4935 Senior Design II	55
Spring 2022	COP 5531 Advanced Algorithm Design and Analysis	8
Fall 2021	COP 4934 Senior Design I	60

Chaibasa Engineering College, India (July 2014 – August 2017)

Course	**Enrollment**
Discrete Mathematics	45
Data Structures	45
Automata	45
Introduction to Computing	90
Computer Organization	45
Algorithms Design and Analysis	45

As a Graduate Teaching Assistant at Auburn University (Fall 2017 – Summer 2021)

COMP 1210 Fundamentals of Computing I
CPSC 1223 Data Structures *(Online)*
COMP 5970/6970 Computational Intelligence & Adversarial Machine Learning
CPSC 1213 Introduction to Computer Science I *(Online)*
CPSC 1223 Introduction to Computer Science II *(Online)*

Course or Curriculum Development (at AUM)

New Courses (Designed from Scratch, starting Fall 2025)

(2) BCMB 6030: Principles of Bioinformatics, 3 Credit Hours: Developed to bridge computational algorithms, including machine learning, NLP, and molecular biology, while emphasizing applications to real-world biological datasets.

(1) Natural Language Processing: Designed to introduce students to core NLP concepts, including large language models (LLMs), with a focus on practical applications.

Significant Revisions / Designed from Scratch

All courses listed below were taught without prior course materials or reference documentation at AUM. I took full responsibility for developing the curriculum, lecture slides, programming assignments, pedagogical strategies, and assessments. My revisions emphasized active learning, career preparation, and interdisciplinary integration.

(6) CSCI 6000: Data Structure & Fund Algorithms Design, 3 Credit Hours: Developed this core graduate-level course independently. Designed lecture slides, created custom LeetCode-style assignments with live demonstrations, implemented scribe note activities to enhance engagement, and organized industry guest sessions with professionals from Google, Intel, and LinkedIn to bridge academic and career preparation.

(5) CSCI 6170: Advanced Network Systems, 3 Credit Hours: Created the graduate course content entirely from scratch, tailored for large class sizes (up to 100 students). Developed original assignments, introduced in-class coding demos, and invited guest speakers from academia and industry to expose students to emerging trends.

(4) CSCI 2200: Discrete Structures, 3 Credit Hours: Taught and developed mid-semester upon departmental need, without prior preparation. Designed all materials independently and introduced Saturday live sessions to accommodate students' preferences. Applied multimedia tools for enhanced delivery.

(3) CSCI 4350: Network Systems, 3 Credit Hours: Revised and delivered with an emphasis on real-world application. Introduced Wireshark-based practical assignments, TCP socket programming in Python, etc. Developed all materials independently.

(2) CSCI 3600: Fundamental Algorithms Design and Analysis, 3 Credit Hours: Designed and implemented the entire course structure. Integrated job-preparatory tools like LeetCode, implemented an active learning model through programming quizzes and discussions, and invited guest speakers from academia and industry to expose students to emerging trends.

(1) CSCI 3400: Data Structures, 3 Credit Hours: Developed the course from scratch, including syllabus, lecture content, and assessments. Introduced in-class coding exercises, real-world analogies, and interactive elements such as scribe notes and peer tutoring. Incorporated LeetCode-style assignments to prepare students for technical interviews.

Master's Theses Supervised

(3) "GoFold: a real-time protein folding program", Kevan Baker, Florida Polytechnic University, Department of Computer Science, Spring 2023. Now: Ph.D. student at Auburn University, AL, USA.

Ongoing

(2) “A new deep learning-based approach for Ortholog Detection”, S. Danwada, AUM. (MS – Fall 2026 expected)

(1) “Embedding-Based Sequence Alignment Using Clustering and Double Dynamic Programming”, Robert Spicer, AUM. (MS – Summer 2026 expected)

Undergraduate Students Supervised

(6) Priscilla Udomprasert (BS – Spring 2026 expected)

(5) William Rochelle (BS – Spring 2026 expected)

(4) Nathaneil Hughes (BS)

(3) Aditya Singh (BS)

(2) Aryavardhan Singh (BS)

(1) Tharalillah Dauda (BS)

Undergraduate Peer Mentor Supervised

(3) Nathaneil Hughes, CSCI 3400, Data Structures, Fall 2023

(2) Om Patel, CSCI 3400, Data Structures, Spring 2023

(1) Riya Patel, CSCI 3600, Fund Algorithms Design and Analysis, Spring 2023

Services

University, College, and Departmental Services

Auburn University at Montgomery

- 2022 - Present: Member, CSE Assistant Professor Search Committee
- 2022 - Present: Member, CSE Lecturer Search Committee
- 2024: Member, CSCI Program Coordinator Search Committee.
- 2023-24: Faculty Member, Instructional Designer Search Committee for College of Sciences

College of Sciences, Auburn University at Montgomery

- 2025: Category Captain Judge, GEARSEF (Greater East Alabama Regional Science and Engineering Fair)
- 2023-24: Faculty Member, Instructional Designer Search Committee for College of Sciences
- 2022: Head Judge, GEARSEF (Greater East Alabama Regional Science and Engineering Fair)

Department of Computer Science, Auburn University at Montgomery

- 2025 - Present: Graduate Student Advisor
- 2022 - Present: Member, Curriculum Committee
- 2022 - Present: Member, Curriculum Quality Control Committee
- 2022 - Present: Member, Assistant Professor Search Committee
- 2022 - Present: Member, Lecturer Search Committee

- 2022 - Present: Member, Grade Dispute Committee
- 2022 - Present: Member, Admission Committee
- 2024: Member, Program Coordinator Search Committee.

Department of Computer Science, Florida Polytechnic University

- 2021 - 22: Lead Organizer of Departmental Research Presentation
- 2021 - 22: Undergraduate Curriculum Committee

Professional

Journal Reviewer

- Information Diffusion *(Impact Factor of 14.8)*
- Bioinformatics *(Impact Factor of 6.9)*
- Public Health Reviews *(Impact Factor of 5.5)*
- Nature Scientific Reports *(Impact Factor of 3.9)*
- Journal of Affective Disorders Reports *(Impact Factor of 4.9)*
- Computational and Structural Biotechnology Reports
- Information Processing and Management *(Impact Factor of 6.9)*

Conference/Workshop Reviewer

• ISMB'25	Intelligent Systems for Molecular Biology
• IEEE-BIBM'25,'19	IEEE International Conference on Bioinformatics and Biomedicine
• ACM/CMD-IT TAPIA'25	Richard Tapia Celebration of Diversity in Computing
• SemEval Workshop'23	International Workshop on Semantic Evaluation

Membership

- 2019, 2023, 2025: MidSouth Computational Biology and Bioinformatics Society (MCBIOS)
- 2022: Intelligent Systems for Molecular Biology

External Academic Service

- 2022-24: External Member, Board of Studies, Department of Computer Science, University Institute of Engineering, Chandigarh University, India.

Outreach

- 2025: Judge at the Alabama Science and Engineering Fair (ASEF) for K12 at Auburn University.
- 2025: Judge, Auburn University Student Research Symposium, Auburn University.
- 2025: Category Captain Judge, GEARSEF (Greater East Alabama Regional Science and Engineering Fair), AUM.
- 2025: Student Scholarship Reviewer at ACM CMD-IT TAPIA Conference.
- 2024: Category Captain Judge at the Alabama Science and Engineering Fair (ASEF)

for K12 at Auburn University.

- 2024: ACM Research Presentation Judge for graduate students at ACM CMD-IT TAPIA Conference.
- 2024: ACM Poster Judge for graduate students at ACM CMD-IT TAPIA Conference.
- 2024: Judge, Graduate Student Research Talk, Auburn University Research Symposium.
- 2023: Judge at the Alabama Science and Engineering Fair (ASEF) for K12 at Auburn University.
- 2023: Head Judge, Greater East Alabama Regional Science and Engineering Fair (GEARSEF) for K12, AUM.
- 2023: Poster Judge at the 19th MCBIOS conference, Dallas.